

Rafaella Sampaio de Alencar

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RESEARCH INTERESTS

Learning Technologies; Personalized or Adaptive Learning Environments; Artificial Intelligence for Education; Human-Computer Interaction; Equity and Technology for the Developing World.

EDUCATION

Ph.D. in Information Science • University of Pittsburgh 2024 - Present

Advisor: Professor Erin Walker

M.S. in Information Science • University of Pittsburgh 2024 - Present

B.S. in Systems and Digital Media • Federal University of Ceará (UFC) 2019 - 2023

Summa Cum Laude

Thesis: A Responsive Web Application to Catalog Public Works

B.S. in Nautical Science • Merchant Marine Officers Training School (EFOMM) 2013 - 2017

Equivalent to a Bachelor's degree in Administration to the Brazilian Federal Council of Administration (CFA)

Thesis: Intro to Remote Sensing: a Tool to Be Explored in the Maritime Navigation

RESEARCH

Graduate Research Student • University of Pittsburgh 01/2026 - Present

Developed and evaluated computational models of word-recall practice for people with aphasia, applying Elo-based prediction algorithms to operationalize adaptive difficulty and personalize rehabilitation practice. This work contributes to the development of a digital therapeutic platform integrating AI-driven individual practice with group-based therapeutic games. Currently preparing IRB materials and co-design sessions with the aphasia community to evaluate and inform the development of the platform prototype.

Resulted in one peer-reviewed conference publication [C.4].

Graduate Research Student • University of Pittsburgh 08/2025 - 12/2025

Contributed to the development and deployment of smart learning content infrastructure for programming education, including maintenance of study environments and collection and organization of research datasets. Led and assisted student modeling work spanning automated generation of knowledge components from coding problems using large language models, alignment of CS1 textbooks and exercises through knowledge component-driven methods, and integration of adaptive learning materials into project-based courses at community colleges.

Resulted in two peer-reviewed conference publications [C.3, C.6] and one peer-reviewed poster [W.1].

Helped planning and executing game-development workshops for high- and middle-school students, with the purpose of teaching them computational skills and increasing their proximity to STEM-related careers. Contributed to IRB preparation for human subjects research. Investigated and developed new tools to aid instructors and students of a hybrid anatomy course in the Physical Therapy department.

Resulted in one peer-reviewed journal publication [J.1].

PUBLICATIONS

PEER-REVIEWED JOURNAL PAPER

- [J.1] Reivian Berrios Barillas, Mariano de Leon, Scott Fraundorf, Peter Brusilovsky, **Rafaella Sampaio de Alencar**, Kim Nixon-Cave, and Yanshan Wang. 2026. [Supplemental AI in a Hybrid Human Anatomy Course and Its Effect on Student Performance and Engagement](#). American Journal of Distance Education, May, 1–13.

PEER-REVIEWED CONFERENCE PAPERS

- [C.6] Zhangqi Duan, Nigel Fernandez, Arun Balajiee Lekshmi Narayanan, Mohammad Hassany, **Rafaella Sampaio de Alencar**, Peter Brusilovsky, Bitu Akram, and Andrew Lan. 2026. [Automated Knowledge Component Generation and Interpretable Knowledge Tracing in Coding Problems](#). In Findings of the Association for Computational Linguistics: ACL 2026, pages 33405–33423, San Diego, California, United States. Association for Computational Linguistics.
- [C.5] **Rafaella Sampaio de Alencar**, Xingyu Song, Daniel Steve Cheng, Jimmara Scott, and Aakash Gautam. 2026. [Cracking the Case, Missing the Data: Design Tensions in Community-Led Critical Data Literacy Instruction](#). In Proceedings of the 2026 Conference on Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT 2026). **Exemplary Paper (Best Paper Nominee)**.
- [C.4] **Rafaella Sampaio de Alencar**, Michael Yudelson, Peter Brusilovsky, and William S. Evans. 2026. [Using Elo to Operationalize Modeling of Aphasia Patients' Word-Recall Practice During Recovery](#). In Proceedings of the 34th ACM Conference on User Modeling, Adaptation and Personalization (UMAP '26).
- [C.3] Mohammad Hassany, Can Kultur, **Rafaella Sampaio de Alencar**, Kamil Akhuseyinoglu, Christopher Bogart, Jaromir Savelka, and Peter Brusilovsky. 2026. [Integrating Smart Learning Content with Project-Based Introductory Programming Course at Community Colleges](#). In Proceedings of the 30th Annual Consortium for Computing Sciences in Colleges - Northeastern Conference (CCSCNE '26).
- [C.2] Rully Agus Hendrawan, **Rafaella Sampaio de Alencar**, Alice Micheli, Peter Brusilovsky, Jordan Barria-Pineda, and Sergey Sosnovsky. 2026. [Data-Driven Evaluation of LLM-Based Ontology Concept Extraction from Programming Learning Content](#). In Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK '26).
- [C.1] **Rafaella Sampaio de Alencar**, Mehmet Arif Demirtas, Aditty Soukarjya Saha, Yang Shi, and Peter Brusilovsky. 2025. [Integrating Expert Knowledge With Automated Knowledge Component Extraction](#)

[for Student Modeling](#). In Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '25). Association for Computing Machinery, New York, NY, USA, 307–312.

PEER-REVIEWED WORKSHOPS PAPER AND POSTER

- [W.2] Rully Agus Hendrawan, Shravya Suresh, **Rafaella Sampaio de Alencar**, Alice Micheli, Peter Brusilovsky and Sergey Sosnovsky. 2026. **Ontology-Guided Knowledge Component Extraction for Programming Exercises**. In Proceedings of the 10th Educational Data Mining in Computer Science Education Workshop (CSEDM'26). Seoul, South Korea. *Proceedings in Progress*.
- [W.1] Samantha Smith, Arun-Balajiee Lekshmi Narayanan, Anurata Prabha Hridi, **Rafaella Sampaio de Alencar**, Bitu Akram, Arto Hellas, Juho Leinonen, Peter Brusilovsky, and Narges Norouzi. 2026. [Knowledge Component-Driven Alignment of CS1 Textbooks and Exercises](#). In Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE TS 2026).

FELLOWSHIPS AND AWARDS

- K. Leroy Irvis Fellowship - 2024-2029
- Discipline-Based Science Education Research Center (dB-SERC) Course Transformation Award - 2025

EMPLOYMENT

Software Developer • Intmed, Brazil

03/2023 - 11/2023

Developed a centralized system for Brazil's largest health maintenance organization, including requests and results of exams, monitoring of patients' vital signs, check-in and check-out of patients, managing beds, etc. The system covered all the organization's hospitals and the different levels of access of different professionals (e.g. physicians, nurses, administrators).

Software Developer • Fundação ASTEF, Brazil

08/2021 - 02/2023

Developed an interactive map of criminal activities for the state's Secretariat of Public Security. This system included several different filters (e.g. types of crimes, specific days of the week or times of the day, timeframes, areas in the city) and personalizations to allow the planning of police operations.

Contributed to both front- and back-end development of a Dell Technologies' world-wide system for support coverage of technical support.

Software Developer • WDA Tecnologia, Brazil

11/2020 - 08/2021

Front-end development of a delivery system and a robust system for requisite control of an electrical company's products.

Proofreader (QA) • Sistema Ari de Sá, Brazil

04/2019 - 09/2022

Review and quality control of the publisher's textbooks, from elementary to high school.

Deck Cadet • Maersk Supply Service

04/2016 - 06/2017

Navigation, administration and communication responsibilities. Intermediation between Brazilian and international crew members. Bilingual presentation of weekly safety meetings.

TEACHING

<i>Teaching Assistant • Japanese</i>	<i>2018 - 2019</i>
<i>Private Tutor • Mathematics, Physics</i>	<i>2018 - 2019</i>
<i>Volunteer Teaching • Mathematics</i>	<i>2014 - 2015</i>

SKILLS

- **Programming/Scripting Languages** - JavaScript, TypeScript, Python, Java, C#, SQL.
- **Frameworks/Tools** - Vue, React, Node, Git.
- **Languages** - Fluent English, native Portuguese, basic/intermediate Japanese, and basic Spanish.

SERVICE

- **Student Volunteer** - 25th International Conference on Artificial Intelligence in Education (AIED 2024)
- **Student Volunteer** - ACM's Technical Symposium on Computer Science Education (SIGCSE TS 2025)
- **Programming Officer** - Latin American Graduate Association of Students (LAGOS) at the University of Pittsburgh (2025 - Present)
- **PC (Program Committee) Member** - Review of papers and posters for ACM's Technical Symposium on Computer Science Education (SIGCSE TS 2026)

PROFESSIONAL DEVELOPMENT

- CMU LearnLab's Summer School 2024